

Gated Context Memory

Paper A Research Brief

Mingjia Shi

Long-Context Memory

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Research question and method

Question

When can an agent answer from a short learned state, and when should it return to raw evidence?

- ① Recurrently encode context chunks into soft memory while carrying earlier memory forward.
- ② Read memory through a small LoRA adapter on the same frozen language model.
- ③ Keep an adapter-free bounded raw path on the same backbone.
- ④ Use held-out confidence to study routing between the two paths.

Positioning

Compression ratio alone is not the contribution. Paper A studies the complete **compact-memory versus raw-evidence interface**, including paired failure, coverage, and cost.

Core result: useful compact state, clear boundary

base	method	QuALITY	BFCL	HotpotQA
Qwen3-8B	Raw ref.	– [†]	92.4	53.7
	SFT ref.	– [†]	95.4	68.8
	LLMLingua-2	– [†]	70.3	22.1
	Compressor (w/o gate)	– [†]	72.3	28.9
	Compressor (w/ gate)	– [†]	88.5	50.9
Qwen3.5-9B	Raw ref.	– [†]	84.5	53.9
	SFT ref.	– [†]	94.9	71.7
	LLMLingua-2	– [†]	60.8	28.9
	Compressor (w/o gate)	– [†]	72.0	30.5
	Compressor (w/ gate)	– [†]	80.5	51.7

- [†]QuALITY is quarantined; BFCL and HotpotQA remain valid.
- SFT is a full-cost adaptation reference, not a compression competitor.
- Formal testing certifies no nonzero coverage; routing is reported as held-out empirical analysis.

Real long-context transfer

base	method	LongBench-v2	∞ Bench-choice	BABILong
Qwen3-8B	Raw ref.	31.2	52.8	18.1
	Compressor (w/o gate)	— [†]	— [†]	— [‡]
	Compressor (w/ gate)	— [†]	— [†]	18.1
Qwen3.5-9B	Raw ref.	33.0	52.4	10.5
	Compressor (w/o gate)	— [†]	— [†]	— [‡]
	Compressor (w/ gate)	— [†]	— [†]	11.9

Task-structured operating region

The compressor without the gate can beat truncated raw on multi-hop LongBench transfer, while raw wins on single-document / extractive targets. The gate selects between the two paths: **Compressor (w/ gate) $\geq$ Raw on every harvested two-base target.**

[†]QuALITY-source invalid; [‡]BABILong collapse. RULER is in an OOM loop.

Execution status and next steps

stage	done	run	fail	pending
core main	60 valid	0	28 invalid	0
replicate + SFT audit	0 valid	0	9 invalid	0
transfer adapters	27 valid	0	16 invalid	0
real long-context	81 valid	0	25 affected	0
generality	13 valid	2 invalid	8	0
budget / RULER	0 valid	1 window	2 fail	18
ablation	0	0	0	36
official baselines	0	0	0	52

Current workers

- 2 stale-loader xLAM QuALITY runs;
- RULER window-8192 is running;
- primary GPU 1 is contended.

Order to completion

- stop stale/OOM workers;
- sync fixes and validate;
- rerun quarantined cells;
- repair RULER/K32;
- ablation, cost, baselines.